

CAPSTONE PROJECT CRICKET WIN PREDICTION

[Document subtitle]

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Capstone Project

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Problem Statement

BCCI has hired an external analytics consulting firm for data analytics. The major objective of this tie up is to extract actionable insights from the historical match data and make strategic changes to make India win. Primary objective is to create Machine Learning models which correctly predicts a win for the Indian Cricket Team. Once a model is developed then you must extract actionable insights and recommendation. Also, below are the details of the next 10 matches, India is going to play. You must predict the result of the matches and if you are getting prediction as a Loss then suggest some changes and re-run your model again until you are getting Win as a prediction. You cannot use the same strategy in the entire series because opponent will get to know your strategy and they can come with counter strategy. Hence for all the below 5 matches you must suggest unique strategies to make India win. The suggestions should be in-line with the variables that have been mentioned in the given data set. Do consider the feasibility of the suggestions very carefully as well.

1. 1 Test match with England in England. All the matches are day matches. In England, it will be rainy season at the time to match.

2. 2 T20 match with Australia in India. All the matches are Day and Night matches. In India, it will be winter season at the time to match.

3. 2 ODI match with Sri Lanka in India. All the matches are Day and Night matches. In India, it will be winter season at the time to match.

Need of the project / Motto

The aim of this study is to develop machine learning models that can predict the outcome of cricket matches for the BCCI, specifically victories for the Indian Cricket Team. We are hoping to train the models on historical match data. Once the models are developed, the goal is to extract actionable insights from them. These insights will be used to make strategic changes or take pivotal decisions to improve the team's performance and increase their chances of winning. This is a data-driven approach to enhance the strategic decision-making process in cricket.

Understanding business and social Opportunity

The business and social opportunity in this project lies in the potential to revolutionize the way cricket matches are strategized and played. By leveraging data analytics and machine learning, the Board of Control for Cricket in India (BCCI) aims to gain a competitive edge in the sport.

This also introduces another unbiased factor when choosing players based on their merits and past data.

From a **business perspective**, this project could lead to more victories for the Indian Cricket Team, enhancing the team's reputation and potentially leading to increased revenue from sponsorships, merchandise sales, and broadcasting rights. It also opens opportunities for collaborations with tech companies and analytics firms, fostering innovation in sports analytics.

From a **social perspective**, cricket is more than just a sport in India; it is a part of the country's culture and identity. Improving the performance of the Indian Cricket Team can boost national pride and unity. Also, we can consistently be better at winning. Additionally, the use of advanced technology in popular sports like cricket can inspire younger generations to develop an interest in cricket as well as studying the sciences.

This could introduce significant changes towards how we look at cricket and would set a precedent for other sports to follow.

Data Report

2929 game details including all three formats of the sport (T20, Test & ODI)

the dataset has 23 variables. Out of which 'Result' is the target variable.

the variables are of 3 different data types of namely object , float64 & int64 respectively.

The dataset has undergone some pretreatment for the missing values and to iron out other errors found in the set. Replacing the string "Three" with the integer 3 in two specific columns. Replacing the string "20-20" with "T20" in the **Match_format** column

Filling missing values in the Match_light_type column with the most frequent value (mode)

Filling missing values in several other columns (Match_format, First_selection, Opponent, Offshore, Season) with their respective most frequent values (mode)

Methodology

The methodology employed for this report includes steps like

- Introduction of the business problem
- Defining problem statement
- Need of the study/project
- Understanding business/social opportunity
- Understanding how data was collected in terms of time, frequency and methodolog
- Visual inspection of data (rows, columns, descriptive details)
- Understanding of attributes (variable info, renaming if required)
- Exploratory data analysis
- Business insights from EDA
- Model Selection & Building
- Final model selection
- Business Insights

Data Inspection

	Game_number	Result	Avg_team_Age	Match_light_type	Match_format	Bowlers_in_team	Wicket_keeper_in_team	All_rounder_in_team	First_selection	Opponent	...	Max_run_scored_1over	Max
0	Game_1	Loss	18.0	Day	ODI	3.0	1	3.0	Bowling	Srilanka	...	13.0	
1	Game_2	Win	24.0	Day	T20	3.0	1	4.0	Batting	Zimbabwe	...	12.0	
2	Game_3	Loss	24.0	Day and Night	T20	3.0	1	2.0	Bowling	Zimbabwe	...	14.0	
3	Game_4	Win	24.0	NaN	ODI	2.0	1	2.0	Bowling	Kenya	...	15.0	
4	Game_5	Loss	24.0	Night	ODI	1.0	1	3.0	Bowling	Srilanka	...	12.0	

5 rows × 23 columns

Fig – 1 Sample of the Dataset

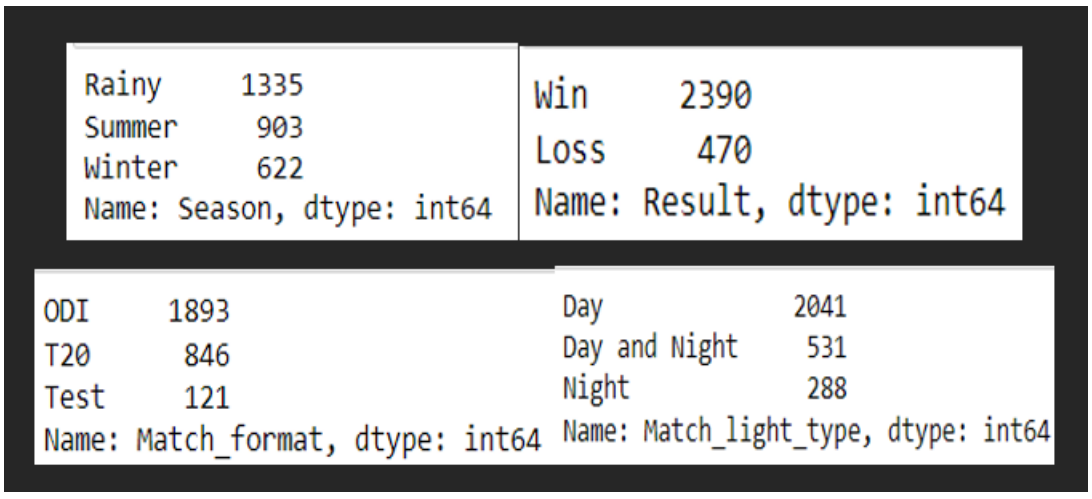


Fig – 2 Value counts of variables

Exploratory Data Analysis

Univariate Analysis

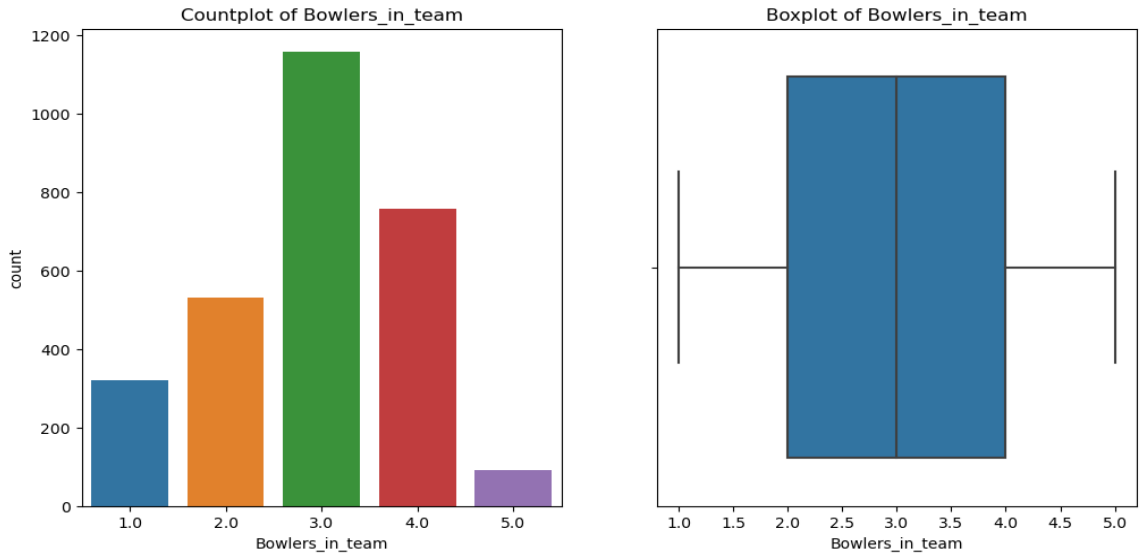


Fig - 3 plots of bowlers in team Vs count

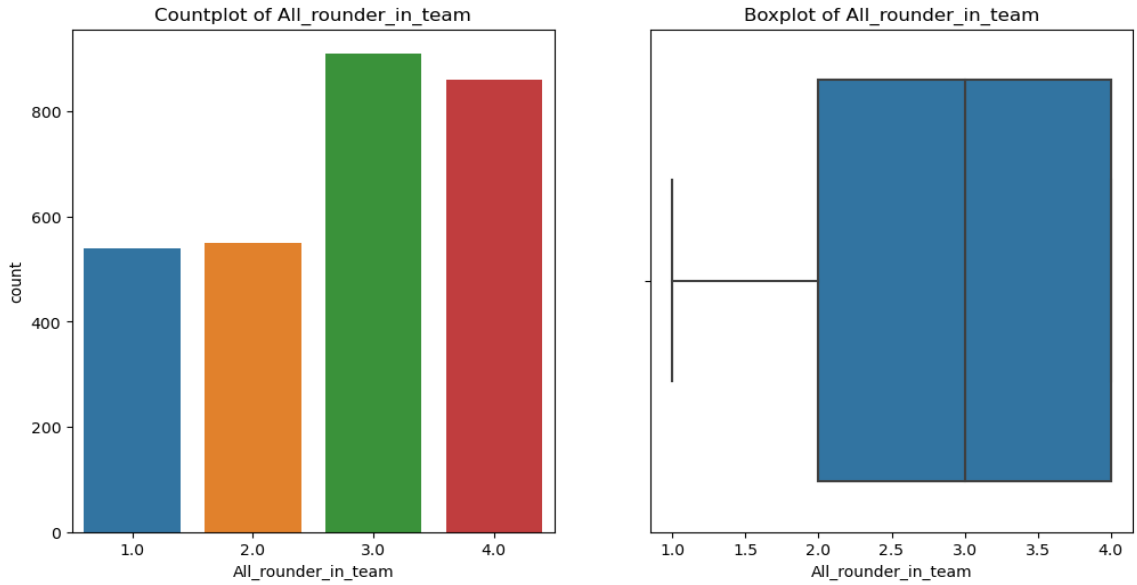


Fig – 4 Plots of All-rounders on the team Vs Count

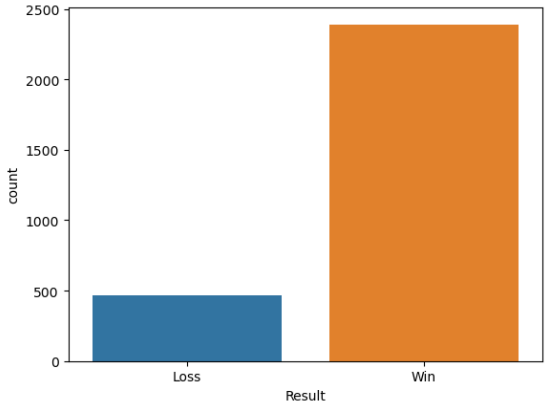


Fig – 5 Countplot of Win Vs Loss

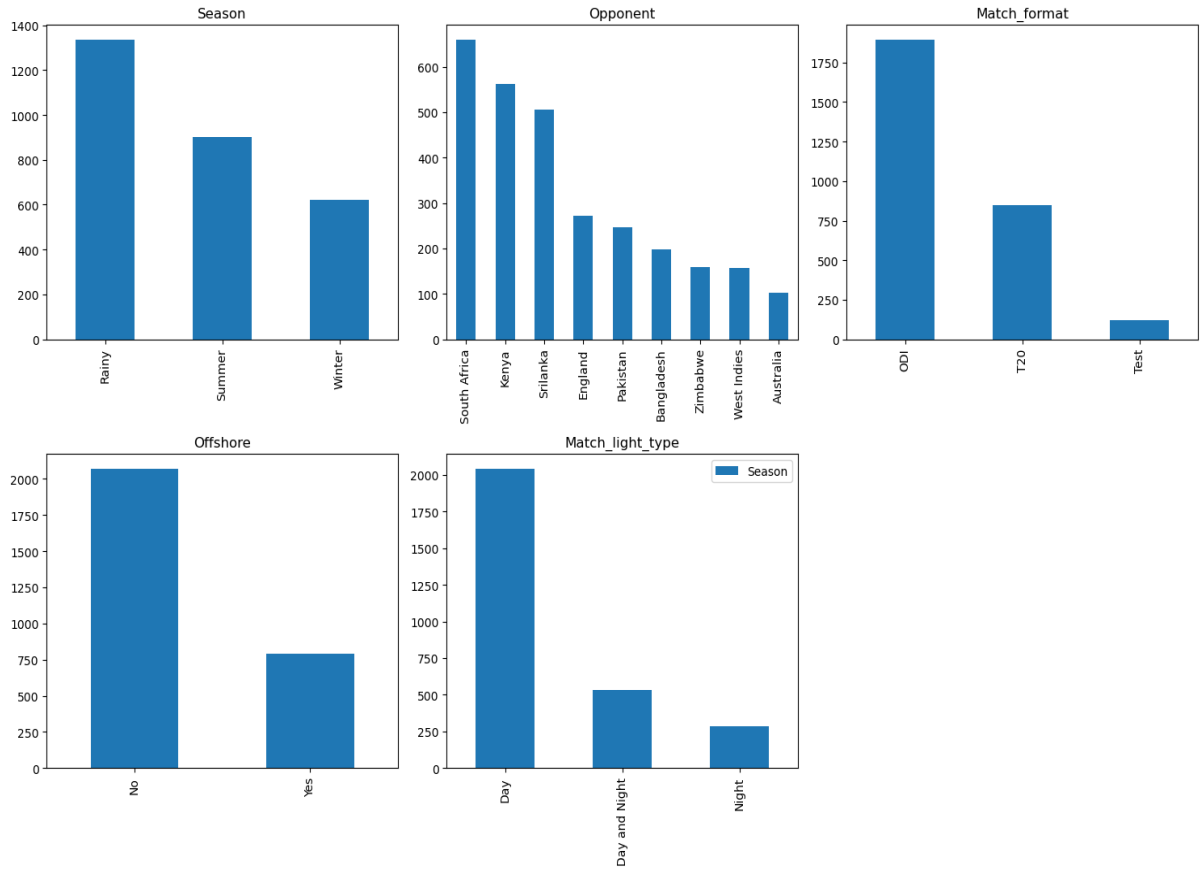


Fig – 6 Multi variable Graph

1. Season: The graph suggests differences in the number of matches played across different seasons. We have played the most matches in the Rainy season (1335) followed by Summer (903) and Winter (622).

2. Opponent: The graph shows the frequency of matches played against various opponents. have played the most matches against South Africa, totaling 659. Kenya follows as the second most encountered team, with India playing 563 matches against them. Sri Lanka is also a frequent opponent, with 505 matches played. The number of matches played against England is 272, while Pakistan has been an opponent in 247 matches. Bangladesh has faced India in 197 matches. The count goes down to 158 matches against Zimbabwe and 157 against West Indies. Lastly, Australia has played 102 matches against India.

3. Match format: ODI (One Day International): India has played 1893 matches.

T20 (Twenty20): India has played 846 matches.

Test: India has played 121 matches.

4. Offshore: A little over 2000 matches were played on shore in India and around 750 were played outside India in other competing countries. This shows that we can play more matches abroad to broaden our horizons and get more experience as well.

Insights

The data suggests a strong preference for scheduling matches during the day. This could be due to various factors such as better visibility, traditional playing hours, or climatic conditions that Favor Day matches. Spectator count and preference

The relatively lower number of day and night matches might indicate limitations in stadium infrastructure for lighting or perhaps preferences of broadcasters and spectators. The timing of matches can also be influenced by considerations of viewer engagement, with day matches possibly attracting larger crowds or television audiences in certain regions.

- Popularity of Formats: The chart suggests that ODIs are the most popular format, likely due to their one-day duration which makes them a good fit for both broadcasters and spectators.

- Commercial Viability: T20s, while less frequent than ODIs, are still quite popular, possibly due to their commercial viability and the excitement they bring to the sport. And the money they generate.

- Traditional Format: The lower frequency of Test matches might reflect a modern preference for shorter, more intense cricket games, although Test matches are still highly regarded by purists for their strategic depth and the true test of skill they provide.

India has played a total of 2860 cricket matches against the listed teams. The team against which India has played the most is South Africa, with 659 matches, indicating a significant cricketing relationship. This is followed by Kenya, with 563 matches, which is notably high and could suggest a strong cricketing engagement between India and Kenya, possibly due to shared cricketing events or developmental tours.

- On the other end of the spectrum, India has played the fewest matches against Australia, with only 102 matches. Considering Australia's prominence in international cricket, this lower number might be influenced by factors such as scheduling conflicts, tournament structures, or the frequency of bilateral series.
- The preference for 3 bowlers could be due to a balanced team strategy, allowing for a mix of batting and bowling options.
- The lack of variation beyond the IQR suggests a stable selection policy, with little experimentation in the number of bowlers.
- This consistency might reflect the team's confidence in their core bowling lineup and their adaptability to different match conditions with a standard number of bowlers.
- Strategic Choice: The preference for bowling first could be due to various strategic reasons such as taking advantage of the pitch conditions, weather, or the team's confidence in their bowling attack.
- Game Conditions: The decision to bat or bowl first is often influenced by the conditions on the day of the match, including the pitch behavior and weather forecast.
- Tactical Adaptability: The data indicates that while there is a slight preference for bowling, the Indian cricket team is adaptable and makes decisions based on the specific context of each match.

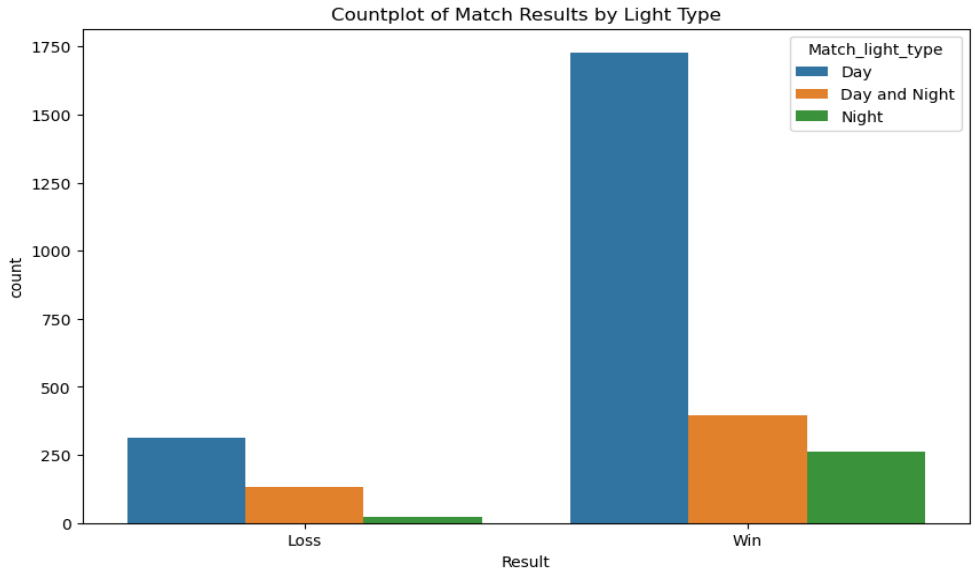


Fig – 7 Plot of Result Vs Match light type

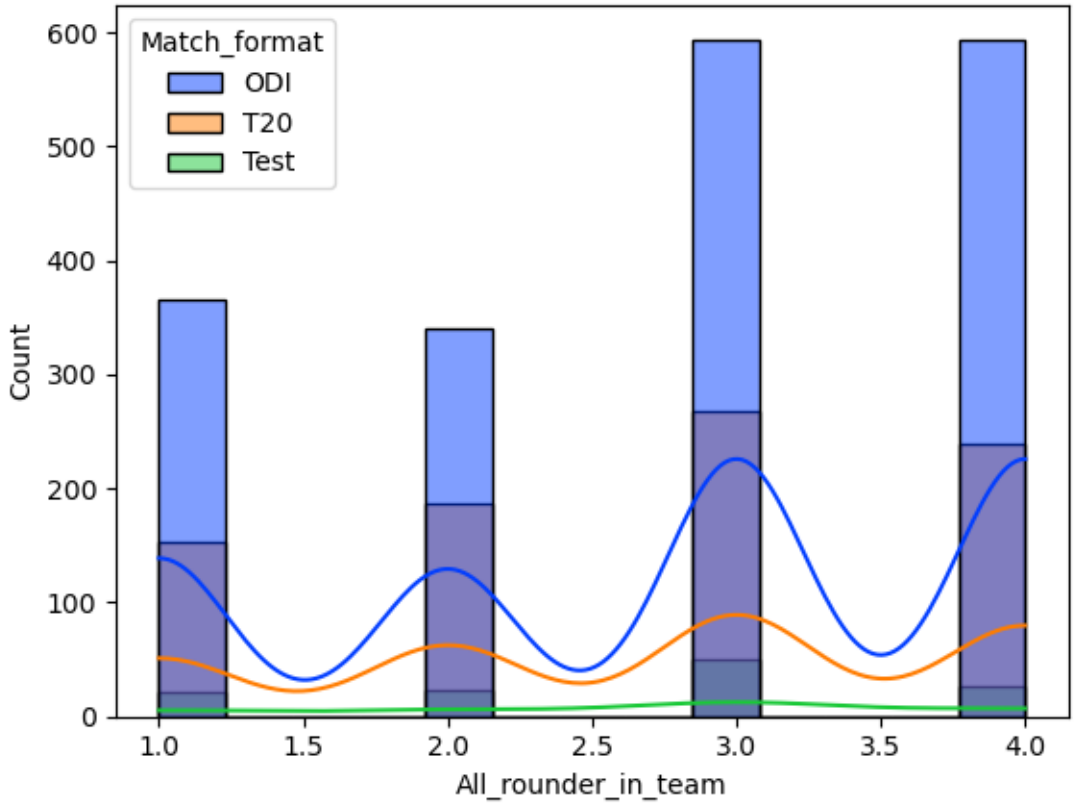


Fig – 8 Plot of All-rounders in team Vs Match Format

Multi variate Analysis

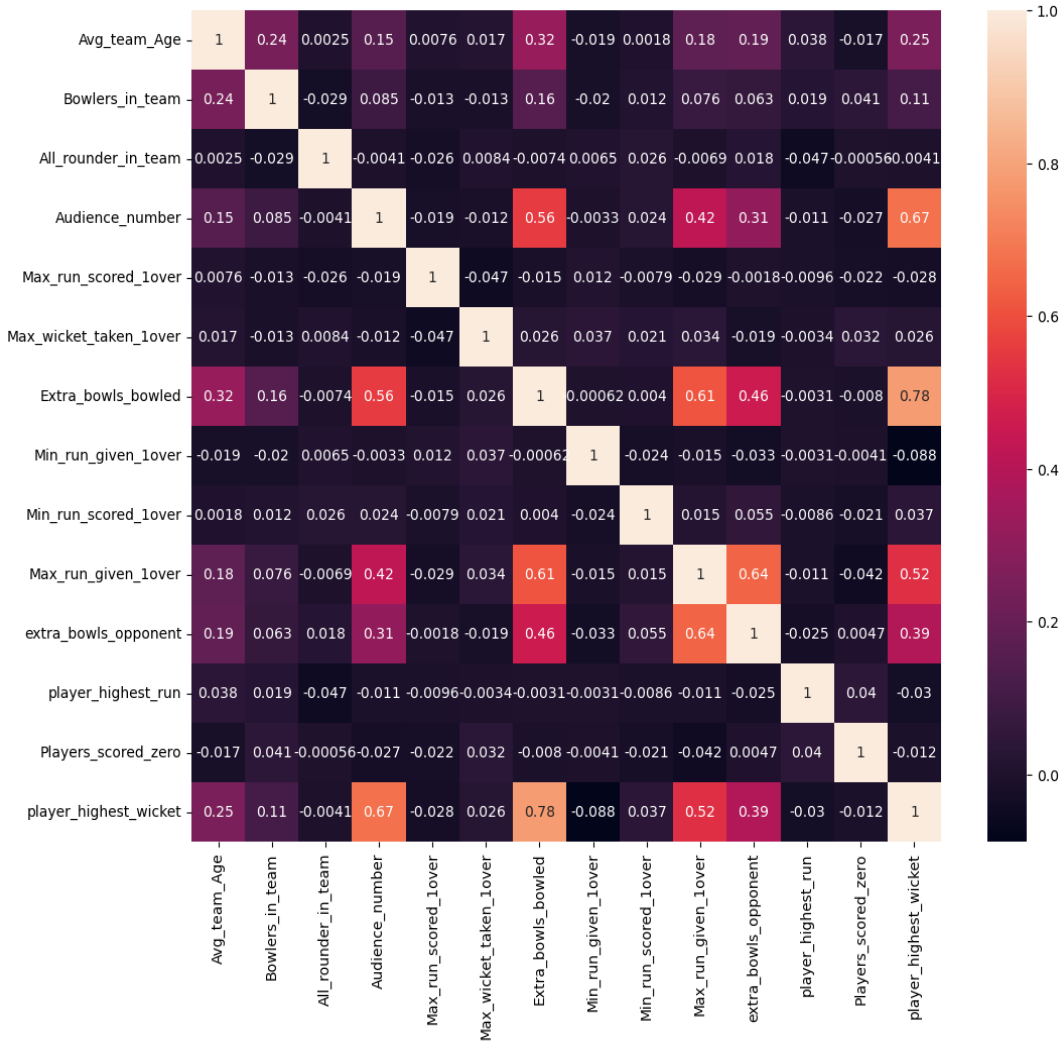


Fig – 9 Multivariate Heat plot

Insights from the Heatmap:

- **Player Performance**: A strong positive correlation between 'Max_run_scored_1over' and 'player_highest_run' suggests that individual player performance in terms of runs scored in an over is a significant contributor to their overall highest run score in the game.
- **Match Outcome**: The correlation between 'Players_scored_zero' and 'Result' indicates that the number of players not scoring runs may affect the match outcome, potentially leading to losses.
- **Team Experience**: A negative correlation between 'Avg_team_Age' and 'player_highest_wicket' could imply that teams with a younger average age may have bowlers taking more wickets, or it might reflect a generational shift in bowling effectiveness.

- **Strategic Decisions**: The relationship between 'First_selection' and other variables like 'Result' or 'Offshore' can provide insights into the strategic decisions made by the team, such as whether choosing to bat or bowl first influences the game's outcome.

Strong Positive Correlations:

Max_run_scored_1over and Player_highest_run: This suggests that in matches where a high number of runs are scored in an over, it's likely that individual players also have high scoring totals. This could indicate that certain overs are particularly pivotal to the outcome of the game and player performance.

Players_scored_zero and Result: A strong correlation here might imply that the number of players who fail to score (scoring zero) can significantly impact the match result, potentially leading to losses.

Extra_bowls_bowled and Max_wicket_taken_1over: This indicates that overs with extra deliveries (like wide balls or no-balls) tend to have more wickets taken. It could suggest that pressure from extra deliveries leads to wickets, or that aggressive bowling that results in extras also leads to dismissals.

Strong Negative Correlations:

Avg_team_Age and Player_highest_wicket: A negative correlation here could suggest that teams with a lower average age may have bowlers who take more wickets. This might reflect the agility and fitness of younger players or a generational shift in bowling talent.

Insights Based on Correlations:

Team Composition: Teams might consider these correlations when selecting players for a match. For example, a younger bowling lineup might be preferred if they are statistically more likely to take wickets.

Match Strategy: The correlation between players scoring zero and match results could lead teams to focus on strategies that minimize the chances of players getting out without scoring.

Training Focus: The positive correlation between extra bowls bowled and wickets taken could be used to emphasize the importance of maintaining pressure during an over, even if extras are bowled.

```
Chi score for Result: 2852.722875863757
Chi score for Avg_team_Age: 137.94008781570903
Chi score for Match_light_type: 45.6708132706161
Chi score for Match_format: 21.233581594465484
Chi score for Bowlers_in_team: 6.1063563405020025
Chi score for All_rounder_in_team: 39.71116580295492
Chi score for First_selection: 2.5578420486314277
Chi score for Opponent: 176.33236522573972
Chi score for Season: 93.9279429970015
Chi score for Audience_number: 2487.264955933411
Chi score for Offshore: 166.8614098581564
Chi score for Max_run_scored_1over: 23.118715986846755
Chi score for Max_wicket_taken_1over: 10.9304962627169
Chi score for Extra_bowls_bowled: 241.7397196208537
Chi score for Min_run_given_1over: 43.82178622279596
Chi score for Min_run_scored_1over: 28.84679372105755
Chi score for Max_run_given_1over: 80.83583662324537
Chi score for extra_bowls_opponent: 125.18736626345965
Chi score for player_highest_run: 139.16360828039507
Chi score for Players_scored_zero: 54.77004511616919
Chi score for player_highest_wicket: 142.01543691554775
```

Fig – 10 Chi Squared scores for variables.

Purpose of Chi-Squared Scores

1. **Feature Selection**:
    - **Identify Important Features**: Chi-squared scores help identify which features are most relevant to the target variable. By calculating the chi-squared statistic for each feature, you can determine the degree of association between the feature and the target variable.
    - **Reduce Dimensionality**: In high-dimensional datasets, not all features contribute equally to the prediction of the target variable. By selecting only the most noteworthy features, you can reduce the dimensionality of the data, leading to simpler and more interpretable models.
2. **Improve Model Performance**:
    - **Avoid Overfitting**: By removing irrelevant or redundant features, you can reduce the risk of overfitting. Models with fewer, more relevant features are less likely to capture noise in the training data.
    - **Enhance Generalization**: Models trained with relevant features are likely to generalize better to unseen data, improving their performance on test datasets.

Model Building

We have built the following models in hopes to better understand this problem. And, to find one model that suits our need to resolve the issue.

1. Logistic Regression
2. K – Nearest Neighbor
3. Random Forest
4. Decision Tree
5. Artificial Neural Networks (ANN)
6. LDA

We have chosen these models as a comprehensive approach to the problem, starting with simpler models like Logistic Regression and moving to more complex ones like ANN. Each of these models has its strengths and weaknesses, and their performance will depend on the specific characteristics of the data. By trying out different models, we can compare their performance and choose the one that works best for your specific problem. As , the best model is not always the most complex one - it is the one that performs best on unseen data.

We tried using the approach of using different models based on the match format, but we realised since the data is so heavily biased the results were the same as when models were employed to all formats together.

Model Selection approach!

1. **Logistic Regression:** Logistic Regression is a simple, interpretable model that is often used as a baseline for classification problems. It is an excellent choice for this problem because it can manage categorical variables.
2. **K-Nearest Neighbour (KNN):** KNN is a non-parametric model that is effective when there are non-linear relationships between features. In cricket, the performance of a team can be influenced by several factors, such as player form, team dynamics, and weather conditions. KNN can capture these complex relationships and provide accurate predictions.
3. **Random Forest:** Random Forest is an ensemble method that combines multiple decision trees to improve accuracy and reduce overfitting. It is a popular choice for classification problems because it can manage high-dimensional data, is robust to outliers, and provides feature importance scores. In cricket, Random Forest can identify the most key factors contributing to a win or loss.
4. **Decision Tree:** Decision Trees are simple, interpretable models that can be used to identify the most prominent features and their interactions. They are useful for identifying the key factors that contribute to a win or loss. Making them an excellent choice for this problem.
5. **Artificial Neural Networks (ANN):** ANNs are powerful models that can learn complex patterns in data. They are particularly useful when there are non-linear relationships between features, which is often the case in cricket. ANNs can learn to recognize patterns in player performance, team strategies, and other factors that influence the outcome of a match.
6. **Linear Discriminant Analysis (LDA):** LDA is a dimensionality reduction technique that can be used for classification problems. It is useful when there are multiple features that are correlated with each other, which is common in cricket data (e.g., batting average, bowling average, and fielding performance). LDA can help identify the most important features and reduce the dimensionality of the data.

MODEL INTERPRETATION

- MODEL PERFORMANCE

Model	Set	Accuracy	Precision	Recall
Logistic Regression	Train	0.64	0.04	0.08
	Train	0.84	1.00	0.91
	Test	0.89	0.06	0.11
	Test	0.84	1.00	0.91
KNN	Train	0.71	0.37	0.49
	Train	0.89	0.97	0.93
	Test	0.48	0.20	0.29
	Test	0.86	0.96	0.90
Decision Tree	Train	1.00	1.00	1.00
	Train	1.00	1.00	1.00
	Test	0.80	0.72	0.76
	Test	0.95	0.97	0.96
Random Forest	Train	1.00	1.00	1.00
	Train	1.00	1.00	1.00
	Test	0.98	0.73	0.84
	Test	0.95	1.00	0.97
LDA	Train	0.65	0.25	0.36
	Train	0.87	0.97	0.92
	Test	0.58	0.22	0.32
	Test	0.86	0.97	0.91
Neural Network	Train	0.17	1.00	0.29
	Train	0.00	0.00	0.00
	Test	0.16	1.00	0.28
	Test	0.00	0.00	0.00

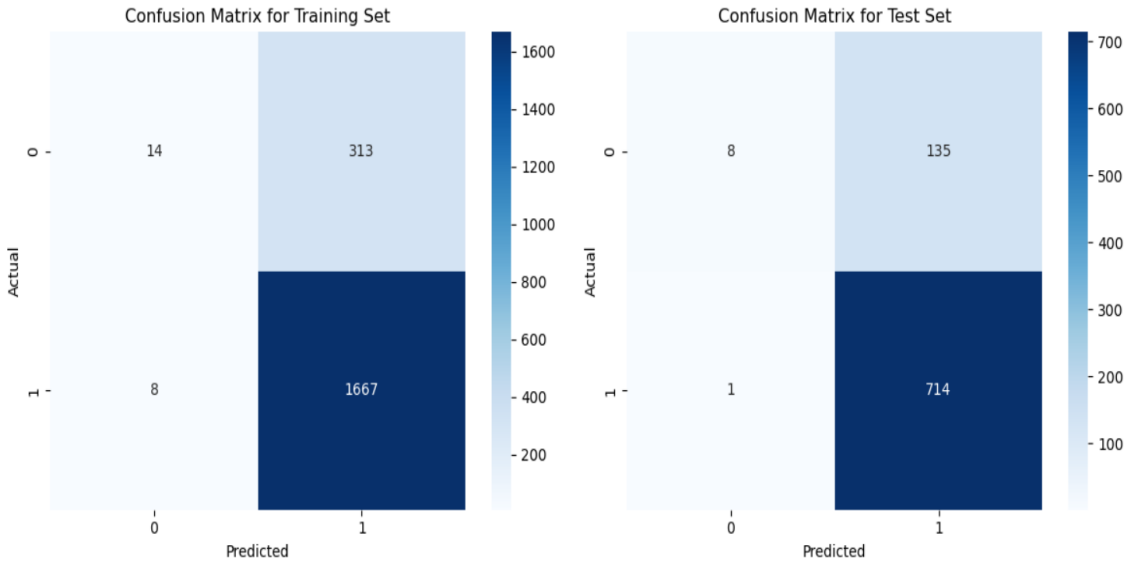


Fig – 11 Confusion matrix for Logistic Regression

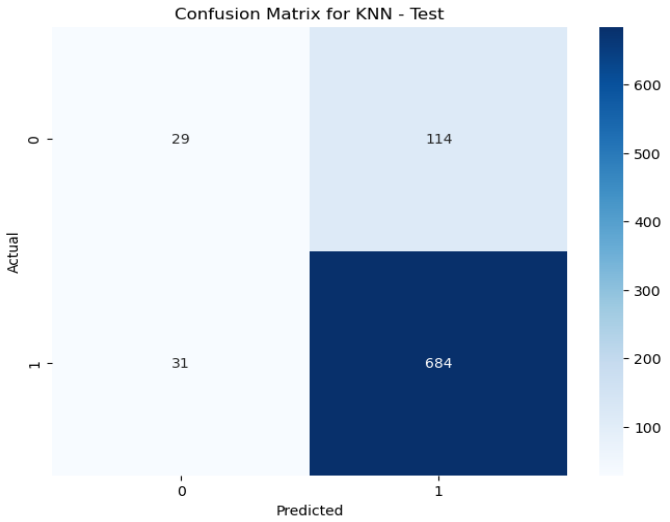


Fig - 12 Confusion Matrix KNN model Test set

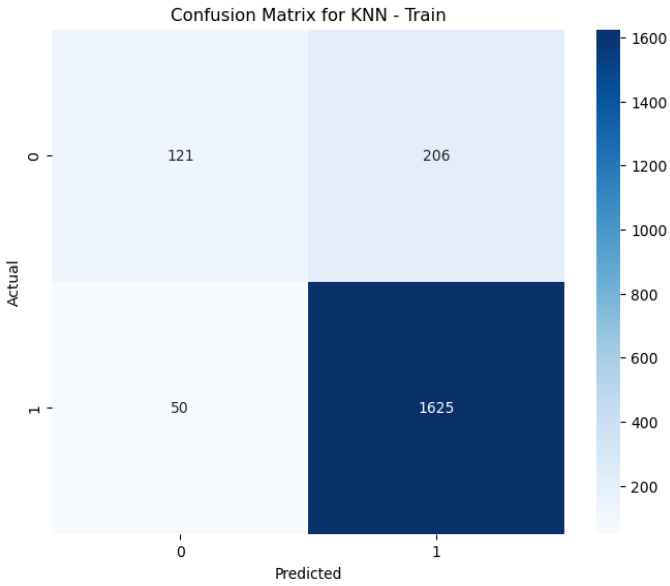


Fig – 13 Confusion Matrix KNN model Train set

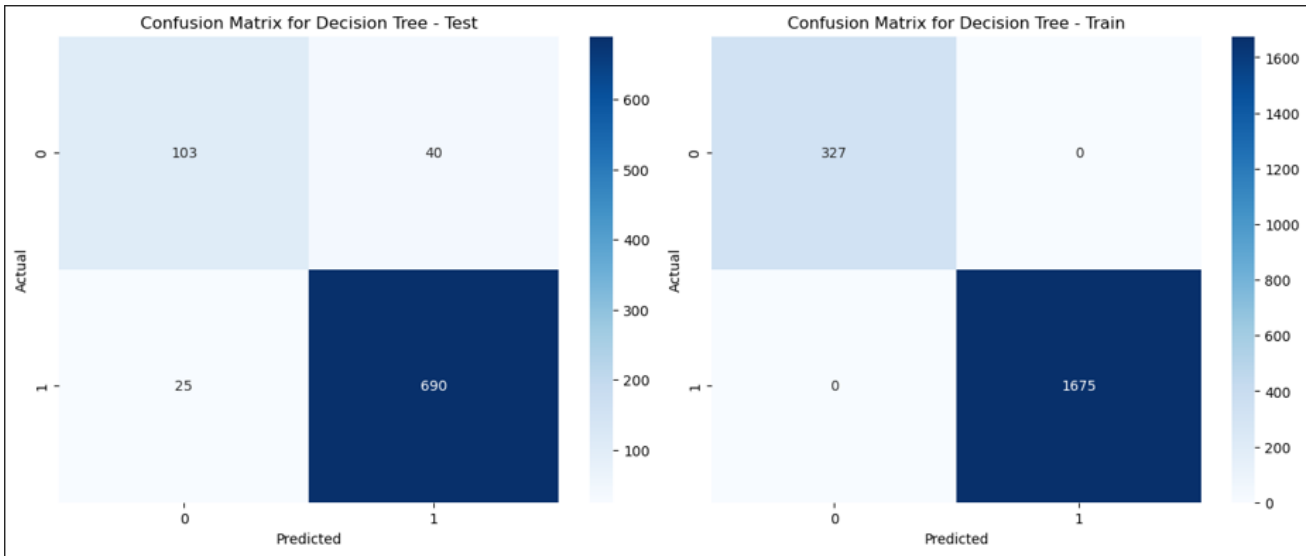


Fig – 14 Confusion matrix Decision Tree model

Random Forest Model

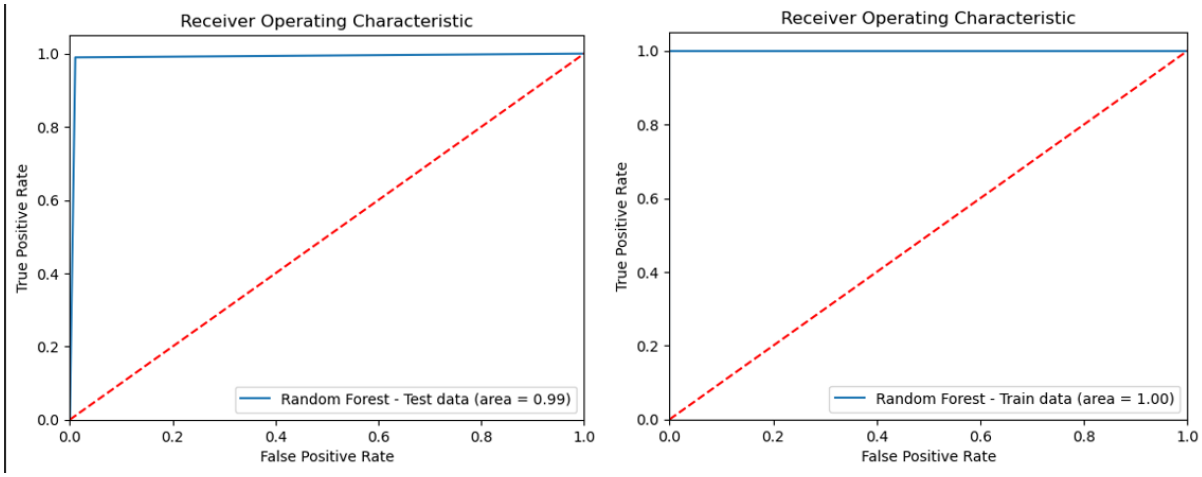


Fig – 15 RF model ROC curve test & train set

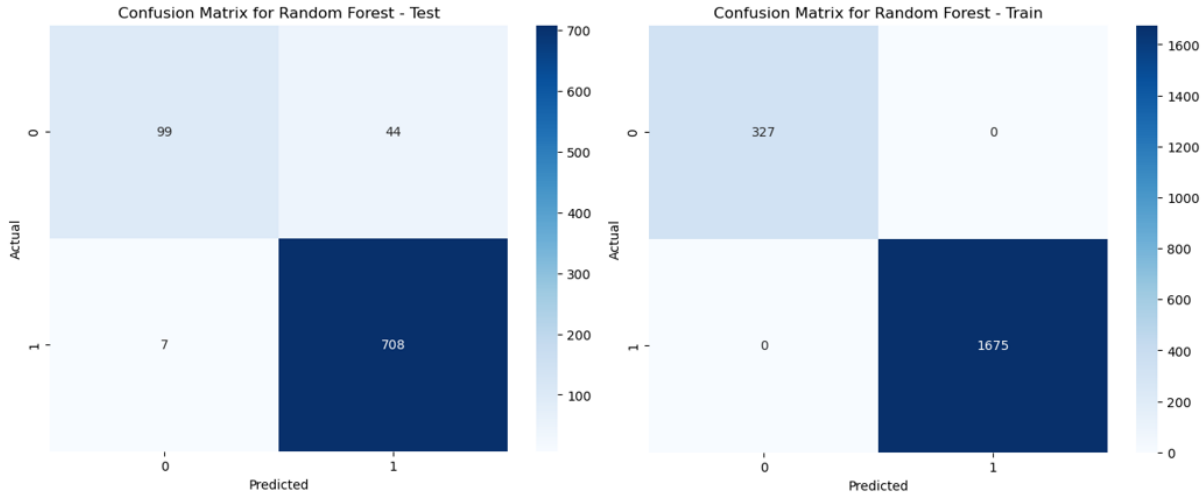


Fig – 15 Confusion matrix Random Forest model

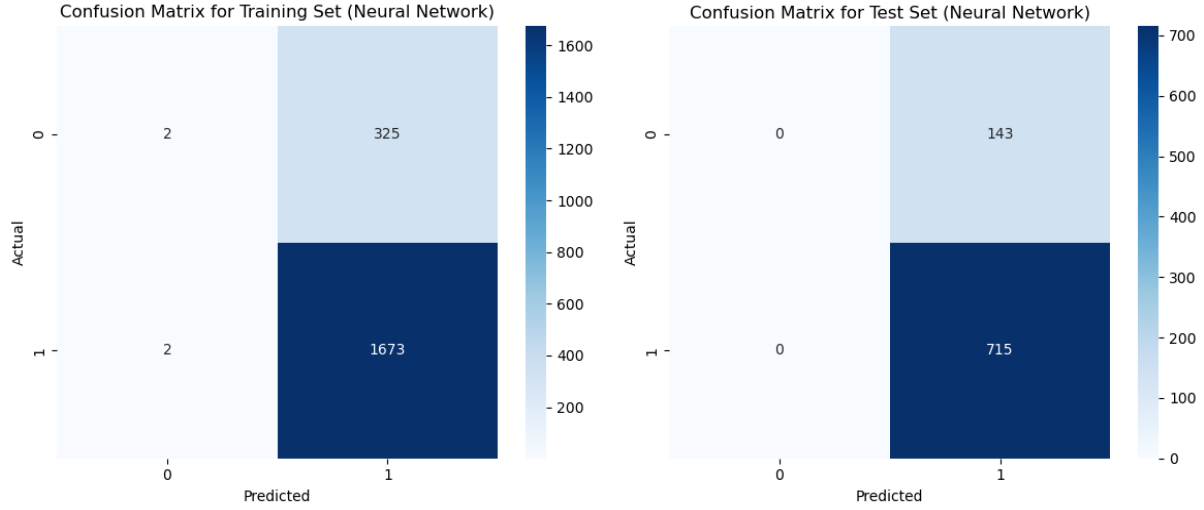
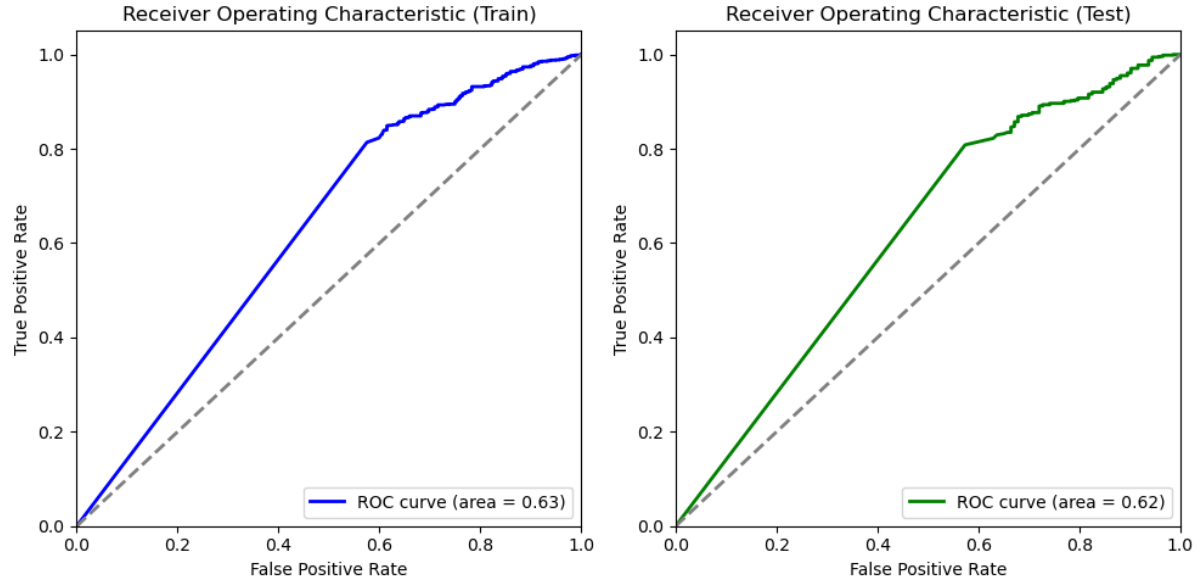


Fig – 16 Performance matrix of ANN model



Insights on Logistic Regression Model

The accuracy of your logistic regression model on both the training and test data is approximately **0.84**. This suggests that the model can correctly predict the target variable about 84% of the time on both the training and test data.

However, the classification reports for both the training and test sets suggest that the model is performing significantly better on predicting class 1 than class 0. This is evident from the high recall of class 1 (**1.00** for both sets) compared to the very low recall of class 0 (**0.04** for the training set and **0.06** for the test set). Recall is the ability of a model to find all the relevant cases within a dataset.

The precision for class 0 is also higher in the test set (**0.89**) compared to the training set (**0.64**). Precision is the ability of a classification model to identify only the relevant data points.

The F1-score, which is the harmonic mean of precision and recall, is also low for class 0 in both sets (**0.08** for the training set and **0.11** for the test set) compared to class 1 (**0.91** for both sets). This indicates that the model is not well balanced and is biased towards class 1.

Insights and Analysis for KNN

- The model's accuracy is high on both the training (0.87) and test (0.83) sets, suggesting that it generalizes reasonably well. However, the significant drop in recall for class 0 from train (0.37) to test (0.20) indicates potential overfitting issues, especially for the minority class (class 0).
- The model performs significantly better on class 1 than on class 0. This is evident in both the training and test sets, where class 1 has higher precision, recall, and F1-scores compared to class 0.
- For the test set, class 0 has a precision of 0.48 and recall of 0.20, indicating that the model struggles to correctly identify instances of class 0. The low recall of 0.20 means that the model misses a lot of actual class 0 instances.
- There is a clear class imbalance, with class 1 having much higher support than class 0 in both the training and test sets. This imbalance contributes to the model's poor performance on class 0, as it has fewer examples to learn from.

Decision Tree Model Insights

Performance on Class 0 vs. Class 1:

- The model performs well on both classes, with higher performance on class 1. For the test set, class 0 has a precision of 0.80 and recall of 0.72, while class 1 has a precision of 0.95 and recall of 0.97. The high recall for class 1 means the model correctly identifies most of the positive cases.

Overfitting:

- The decision tree model has perfect scores (1.00) on the training set for both classes, which is a strong indication of overfitting. The model has learned the training data too well, capturing even the noise.

Generalization:

- Despite overfitting on the training data, the model performs quite well on the test set, achieving an overall accuracy of 0.92. However, the drop in performance from the training set to the test set highlights the overfitting issue, especially for class 0.

Class Imbalance:

- While there is still a class imbalance, the decision tree model handles it better than the KNN model, as evidenced by the high-performance metrics for both classes in the test set.

Key Observations and Insights ANN Model

Model Performance:

The accuracy scores for both the training set (83.67%) and the test set (83.33%) are very close, indicating that the model is performing consistently across both datasets.

However, the ROC curves for both sets show AUC values of 0.63 (training) and 0.62 (test), which are relatively low and indicate that the model's ability to distinguish between classes is limited.

Comparison with Previous Models:

- Compared to the previously analyzed Random Forest model, which had near-perfect AUC scores, the neural network model's AUC scores are significantly lower. This suggests that the neural network model is not as effective in distinguishing between classes as the Random Forest model.
- The ROC curves for the neural network model are closer to the diagonal line (random guessing) than the top-left corner (perfect classification), further emphasizing its weaker performance in terms of class discrimination.
- Despite the lower AUC values, the consistency in accuracy between the training and test sets indicates that the model is not overfitting. The neural network model generalizes well, as evidenced by the similar performance on both datasets.

Insights and Analysis Random Forest model

- Despite overfitting on the training set, the Random Forest model performs excellently on the test set with an overall accuracy of 0.95. The performance metrics are high across the board, demonstrating that the model generalizes well.

- The Random Forest model performs very well on both classes, especially on class 1. For the test set, class 0 has a precision of 0.98 and recall of 0.73, while class 1 has a precision of 0.95 and recall of 1.00. The high precision and recall for class 1 indicate that the model is very effective in identifying positive cases.
- However, the recall for class 0 is lower (0.73), meaning the model misses some actual class 0 instances.
- The ROC curve for the Random Forest model on the training data is essentially a straight line at the top of the plot (TPR = 1.0) and vertical line at the left (FPR = 0.0), which indicates perfect classification.
- The Area Under the Curve (AUC) is 1.00, meaning the model has 100% ability to distinguish between the positive and negative classes on the training set.
- The ROC curve for the Random Forest model on the test data is almost a straight line near the top-left corner of the plot, indicating excellent classification performance.
- The Area Under the Curve (AUC) is 0.99, which suggests the model has very high ability to distinguish between the positive and negative classes on the test set.
- The high AUC score (0.99) on the test set indicates that the model generalizes well to unseen data, performing almost perfectly in distinguishing between classes.
- The slight drop from the perfect AUC of 1.00 on the training set to 0.99 on the test set suggests that while there is a small degree of overfitting, it is not significantly impacting the model's ability to generalize.

Model Robustness:

- Despite the overfitting, the model's performance on the test set is outstanding, with an AUC of 0.99. This demonstrates that the Random Forest model is robust and effective in generalizing from training data to unseen data.

Balanced Performance:

- The high AUC values for both training and test sets indicate that the model has a balanced performance, effectively distinguishing between the positive and negative classes without bias towards either class.

Tuning Models for better results

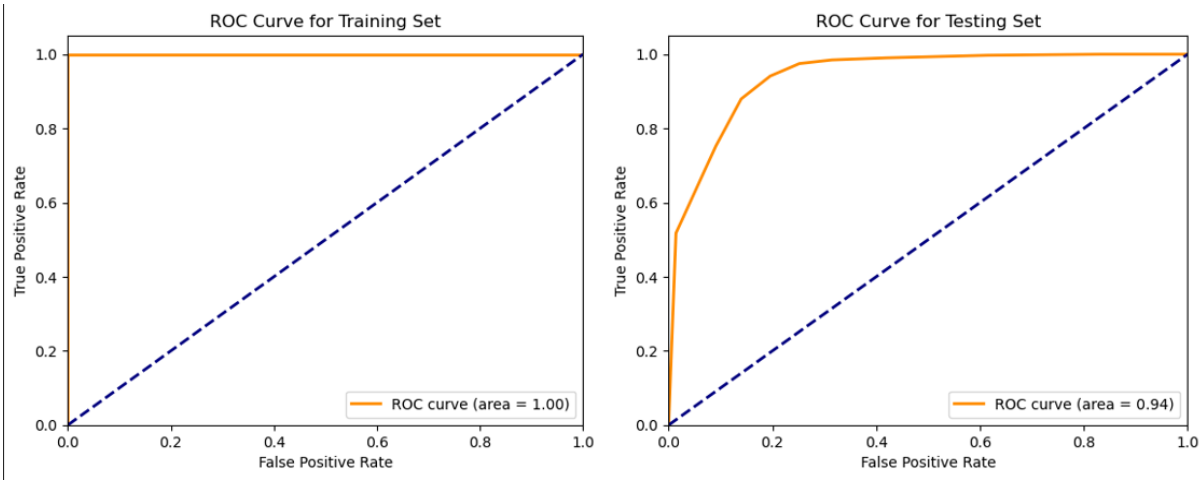


Fig – 17 Bagging Model ROC curve for test & train set

```
Classification Report for Training Set:
              precision    recall  f1-score   support

           0       1.00      1.00      1.00       327
           1       1.00      1.00      1.00      1675

    accuracy                           1.00      2002
   macro avg       1.00      1.00      1.00      2002
weighted avg       1.00      1.00      1.00      2002

Classification Report for Testing Set:
              precision    recall  f1-score   support

           0       0.86      0.75      0.80       143
           1       0.95      0.97      0.96       715

    accuracy                           0.94       858
   macro avg       0.90      0.86      0.88       858
weighted avg       0.94      0.94      0.94       858
```

Fig – 18 Bagging model Performance matrix for test & train set

Summary and Insights

Overfitting on the Training Set: The model performs perfectly on the training set, which indicates that it has likely overfitted to the training data. This is evident from the perfect scores across all metrics.

Despite the overfitting on the training set, the model still performs very well on the testing set with an accuracy of 94%. This suggests that the bagging approach has managed to generalize reasonably well to unseen data.

Class Imbalance Handling:

- Class 0: The precision is high (0.86), but the recall is somewhat lower (0.75). This indicates that the model is good at predicting true negatives but misses some true positives.
- Class 1: The model performs excellently for Class 1, with high precision (0.95) and recall (0.97). This is indicative of the model's effectiveness in correctly identifying the majority class.

Macro and Weighted Averages:

The macro average and weighted average scores provide a balanced view of the model's performance across both classes. The macro averages are slightly lower than the weighted averages, reflecting the impact of the lower recall for Class 0.

SMOTE was also performed to get rid of the imbalance in models and test further.

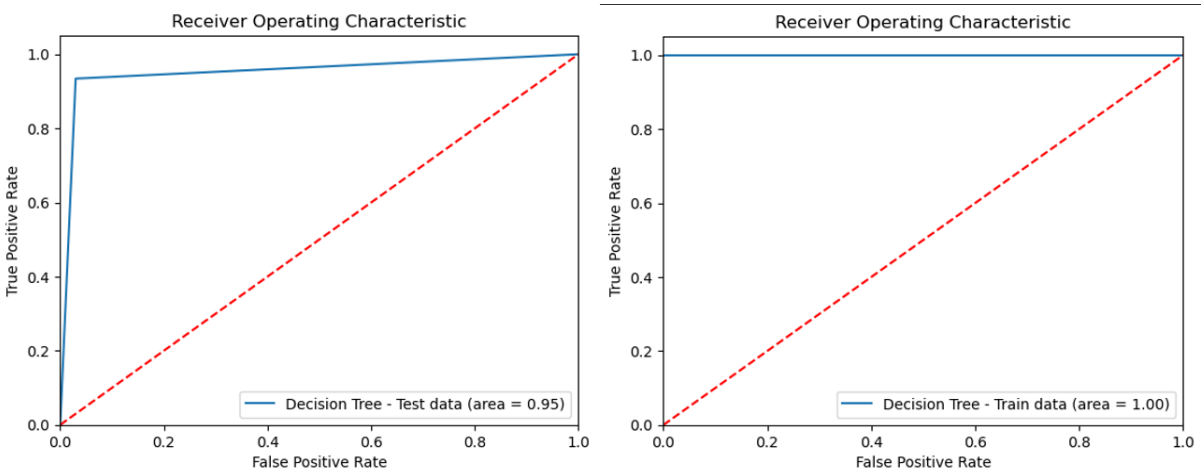


Fig – 19 Decision tree model post SMOTE ROC curve test & train set

Classification Report for Decision Tree - Test data:

	precision	recall	f1-score	support
0	0.93	0.97	0.95	469
1	0.97	0.93	0.95	487
accuracy			0.95	956
macro avg	0.95	0.95	0.95	956
weighted avg	0.95	0.95	0.95	956

Classification Report for Decision Tree - Train data:

	precision	recall	f1-score	support
0	1.00	1.00	1.00	1921
1	1.00	1.00	1.00	1903
accuracy			1.00	3824
macro avg	1.00	1.00	1.00	3824
weighted avg	1.00	1.00	1.00	3824

Fig – 20 performance matrix for Decision tree post SMOTE train & test set

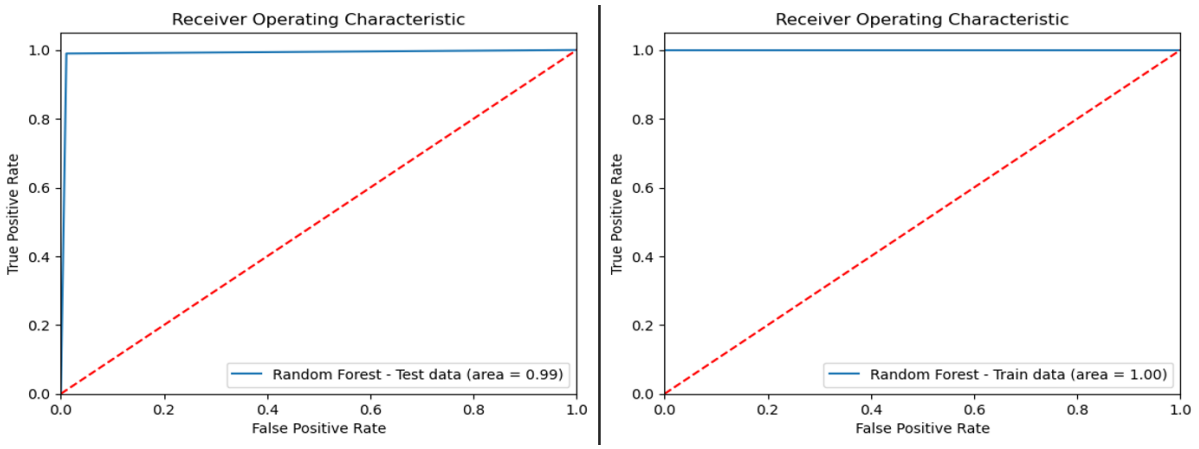


Fig – 21 RF model post SMOTE - ROC curve test & train set

Classification Report for Random Forest - Test data:

	precision	recall	f1-score	support
0	0.99	0.99	0.99	469
1	0.99	0.99	0.99	487
accuracy			0.99	956
macro avg	0.99	0.99	0.99	956
weighted avg	0.99	0.99	0.99	956

Classification Report for Random Forest - Train data:

	precision	recall	f1-score	support
0	1.00	1.00	1.00	1921
1	1.00	1.00	1.00	1903
accuracy			1.00	3824
macro avg	1.00	1.00	1.00	3824
weighted avg	1.00	1.00	1.00	3824

Model Improvement:

After applying SMOTE, the model's performance on the test data has significantly improved. The metrics (precision, recall, and F1-score) for both classes are now balanced and high, indicating that the model is effective in both identifying the minority class (class 1) and maintaining high accuracy for the majority class (class 0).

Class Imbalance Handling:

SMOTE has helped in mitigating the issues caused by class imbalance. The high recall for both classes suggests that the model is now capable of correctly identifying instances of both classes, especially crucial for the minority class where improvement is typically sought after SMOTE treatment.

Precision-Recall :

The model shows a slight trade-off between precision and recall for class 0 versus class 1. Class 1 has higher precision, but slightly lower recall compared to class 0, indicating it's better at avoiding false positives but might miss some true positives compared to class 0.

Overall Performance:

With a weighted average F1-score of 95% and accuracy of 95%, the model performs very well on the test data. It demonstrates robustness in handling unseen data after being trained with SMOTE-enhanced training data.

In summary, the Decision Tree model shows substantial improvement after SMOTE treatment, achieving high accuracy, precision, recall, and F1-score on the test data. This indicates that SMOTE effectively addressed class imbalance, allowing the model to generalize better and make more reliable predictions across both classes.

Insights RF post SMOTE

Excellent Generalization:

The Random Forest model shows outstanding performance on both the training and test data, with all metrics (precision, recall, F1-score, and accuracy) consistently high.

Robustness post-SMOTE:

Like the Decision Tree model, the Random Forest model benefits significantly from the SMOTE treatment. It effectively manages class imbalance, resulting in improved performance metrics across the board.

Precision-Recall Balance:

The model achieves a balanced precision-recall trade-off, with both metrics at 99% for both classes. This balance is crucial for tasks where minimizing false positives (precision) and false negatives (recall) are equally important.

Comparison with Decision Tree:

While both models show excellent performance post-SMOTE, the Random Forest slightly outperforms the Decision Tree, especially in terms of precision and recall on the test data. This suggests that the ensemble approach and averaging of multiple decision trees in Random Forest contribute to better generalization.

Model Reliability:

With weighted averages of 99% for both F1-score and accuracy, the Random Forest model is highly reliable and suitable for deployment in real-world scenarios where high predictive accuracy is essential.

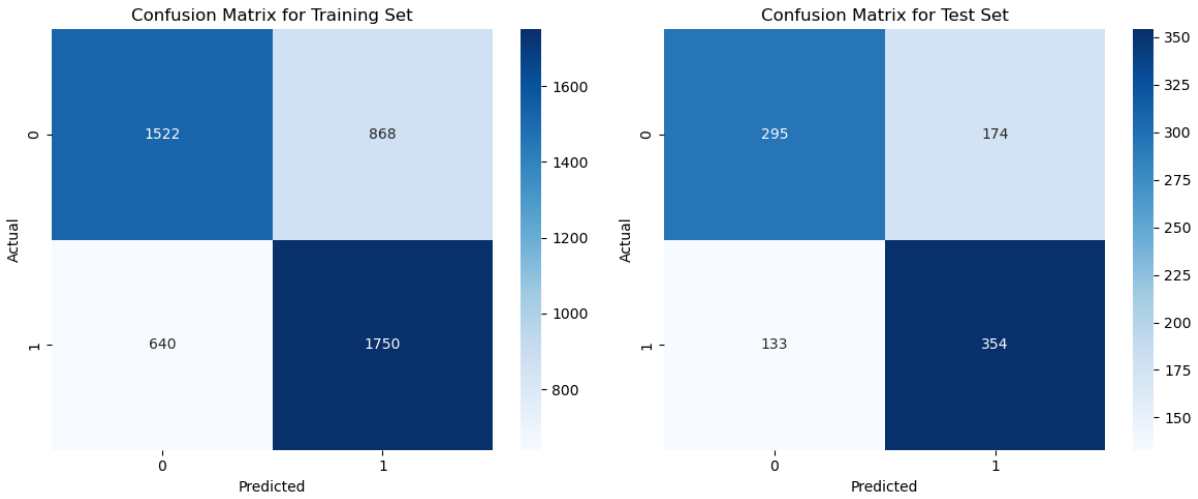


Fig - 22 Logistic Regression heatmap post SMOTE treatment

```
Classification report for the training set:
              precision    recall  f1-score   support

           0       0.70      0.64      0.67      2390
           1       0.67      0.73      0.70      2390

    accuracy                           0.68      4780
   macro avg       0.69      0.68      0.68      4780
weighted avg       0.69      0.68      0.68      4780

Classification report for the test set:
              precision    recall  f1-score   support

           0       0.69      0.63      0.66       469
           1       0.67      0.73      0.70       487

    accuracy                           0.68       956
   macro avg       0.68      0.68      0.68       956
weighted avg       0.68      0.68      0.68       956
```

Fig – 23 Logistic Regression post SMOTE treatment

Logistic regression post SMOTE treatment

Insights:

1. **Performance**:

   The Logistic Regression model, post SMOTE treatment, shows moderate performance on both the training and test data sets. The F1-scores, precision, recall, and accuracy metrics are balanced but not as high as those achieved by Decision Tree or Random Forest models.

2. **Impact of SMOTE**:

   SMOTE has helped in improving the recall for class 0, as seen in the test data, but precision remains moderate. This indicates the model's ability to better capture instances of the minority class (class 1).

3. **Comparison with Ensemble Models**:

   Compared to Decision Tree and Random Forest models, the Logistic Regression model shows lower overall performance metrics (precision, recall, and F1-score).

LDA model pre-SMOTE treatment

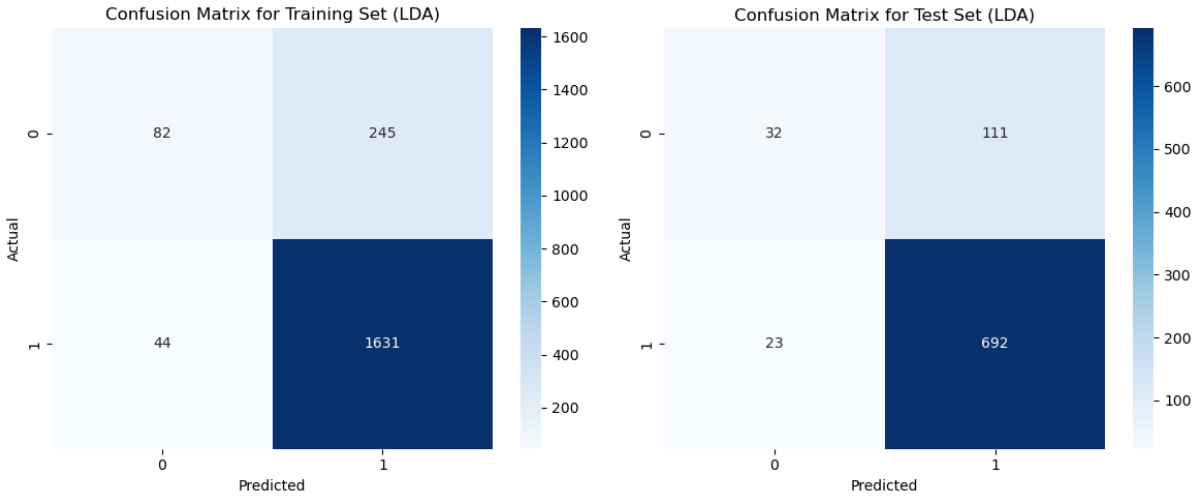


Fig – 24 LDA performance heatmap pre-SMOTE.

Classification Report for Training Set (LDA):

	precision	recall	f1-score	support
0	0.65	0.25	0.36	327
1	0.87	0.97	0.92	1675
accuracy			0.86	2002
macro avg	0.76	0.61	0.64	2002
weighted avg	0.83	0.86	0.83	2002

Classification Report for Test Set (LDA):

	precision	recall	f1-score	support
0	0.58	0.22	0.32	143
1	0.86	0.97	0.91	715
accuracy			0.84	858
macro avg	0.72	0.60	0.62	858
weighted avg	0.82	0.84	0.81	858

Fig – 25 LDA performance matrix pre-SMOTE.

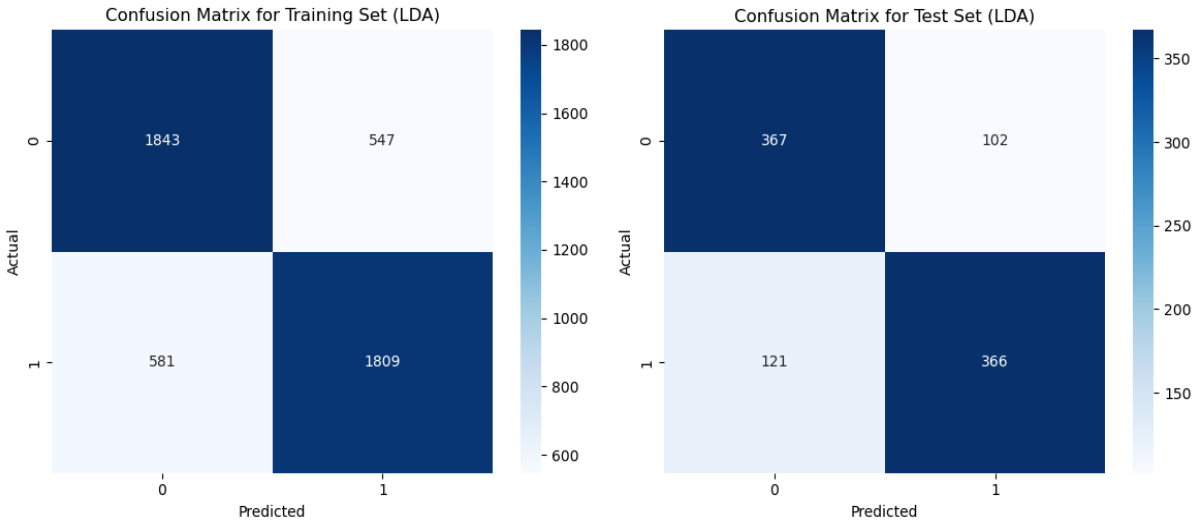


Fig – 26 LDA performance matrix post SMOTE

Training Set: The model had high precision (0.87) and recall (0.97) for class 1 (majority class) but struggled with class 0 (minority class) with low precision (0.65) and recall (0.25). This imbalance resulted in an overall high accuracy of 0.86, driven by the performance on the majority class.

Test Set: Like the training set, the model had good precision and recall for class 1 but significantly lower performance for class 0. The overall accuracy was 0.84, reflecting the imbalanced performance across classes.

Performance After SMOTE:

Training Set: Post-SMOTE, the model's performance metrics (precision, recall, and F1-score) for both classes (0 and 1) became more balanced. While precision and recall decreased slightly compared to the majority class before SMOTE, they improved significantly for the minority class (class 0).

Test Set: The model's performance on the test set also showed improvement in precision and recall for both classes. The F1-scores for both classes became more balanced, resulting in a higher overall accuracy of 0.77 compared to before SMOTE.

Impact of SMOTE on LDA:

SMOTE effectively addressed the class imbalance issue in the dataset by oversampling the minority class (class 0). This resulted in improved model performance metrics for the minority class in both training and test sets.

The balanced performance metrics after SMOTE (higher precision and recall for class 0) indicate that the model is now better equipped to generalize and make accurate predictions across both classes.

Model Selection:

After SMOTE: The LDA model shows more balanced and reliable performance metrics across both training and test sets. It achieves comparable accuracy while ensuring that predictions for both classes are more accurate and balanced.

Conclusion: The LDA model after SMOTE is preferable due to its improved ability to handle class imbalance and provide more reliable predictions across both classes. It strikes a better balance between precision and recall, which is crucial for scenarios where predicting both classes accurately is important.

In summary, the LDA model's performance significantly improves after applying SMOTE, demonstrating enhanced generalization and reliability in predicting outcomes for both classes in the dataset.

Model Comparison:

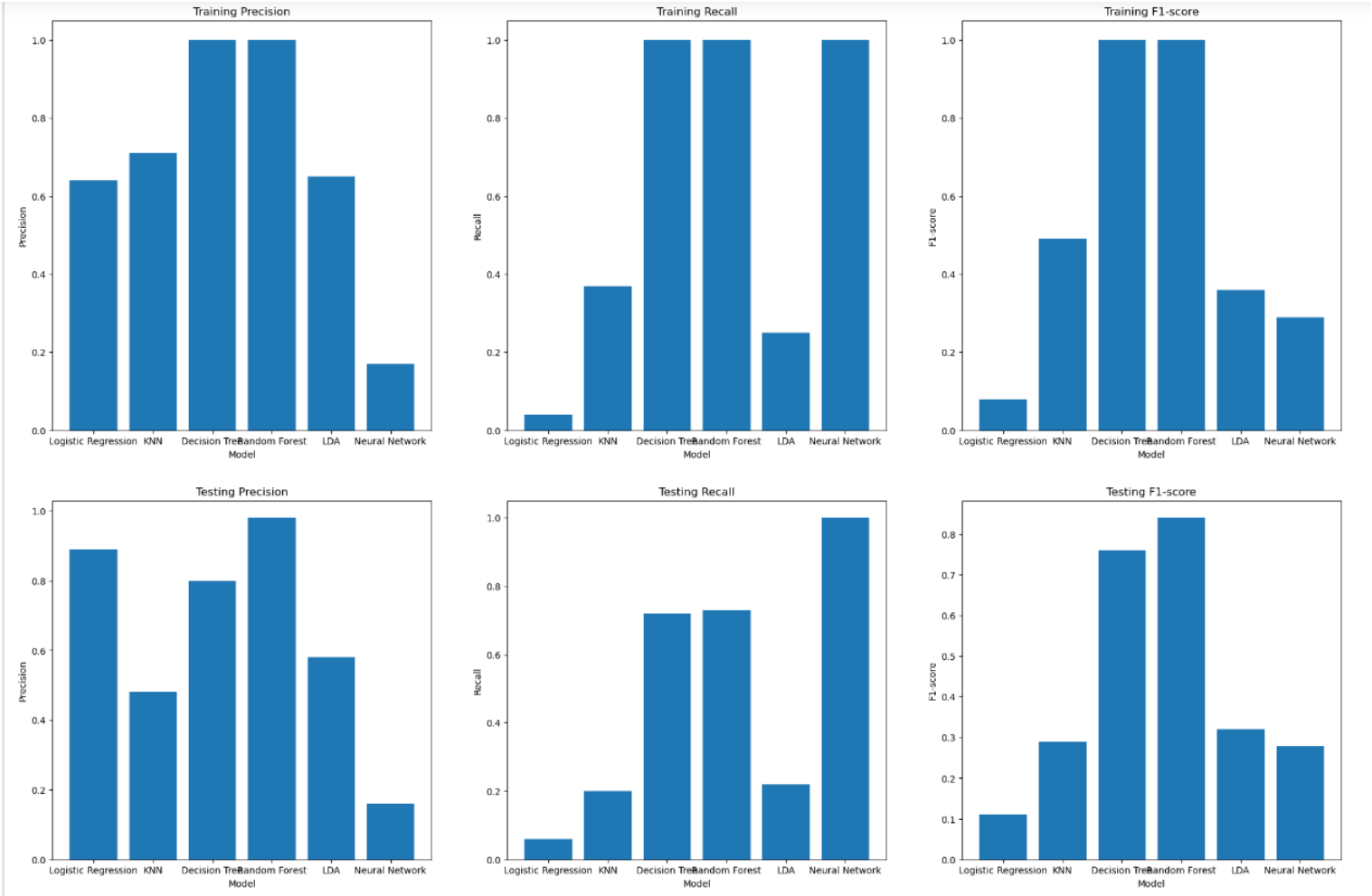


Fig – 27 Model Comparison visual

Insights on Best model and its implications on business

Insights and Recommendations:

- **Accuracy Comparison**: All models show high accuracy, especially after SMOTE treatment. However, accuracy alone may not be the best metric for imbalanced datasets like this one.
- **Precision and Recall**:
    - **Before SMOTE**: Decision Tree and Random Forest had better precision and recall compared to KNN, especially for the minority class (class 0).
    - **After SMOTE**: All models show improved performance metrics for both classes, indicating that SMOTE effectively balanced the predictions.
- **Best Model Selection**:
    - **Decision Tree and Random Forest** perform exceptionally well after SMOTE, with perfect scores on training data and high precision and recall on test data. They are particularly good at handling imbalanced classes and would be reliable for predicting both wins and losses.
    - **Bagging Model (based on Decision Tree)** also shows promising results, slightly below Random Forest but still achieving high accuracy and balanced precision-recall scores.
- **Recommendation**: Based on the analysis, **Random Forest** seems to be the best model for predicting win or loss for the Indian cricket team in the next 5 matches. It consistently shows high accuracy, precision, and recall after SMOTE treatment, making it robust for handling potential imbalances in the dataset. Its ensemble nature and ability to capture complex relationships in the data also contribute to its suitability for this prediction task.

*Therefore, **Random Forest** is recommended as the model of choice for predicting outcomes in the upcoming matches, ensuring reliable and accurate predictions based on the historical data provided.*

Chi score & Model based Insights and Recommendations

The Chi-square scores indicate the strength of the relationship between each variable and the match outcome. Higher scores suggest a stronger relationship. Here are some insights based on the scores:

1. **Result (2852.72)** and **Audience_number (2487.26)**: These variables have the highest Chi-square scores, indicating they have the strongest relationships with the match outcome. However, these aren't variables that can be controlled or adjusted.
2. **Avg_team_Age (137.94)**, **player_highest_run (139.16)**, and **player_highest_wicket (142.01)**: These variables suggest that the performance of individual players and the average age of the team are important factors. To improve the chances of winning, the team could focus on player training and fitness, particularly for key players who have the potential to score high runs or take many wickets. If the team's average age is high, introducing some younger players might bring in new energy and perspectives.
3. **Opponent (176.33)** and **Offshore (166.86)**: The opponent team and whether the match is played at home or offshore (abroad) also seem to significantly affect the match outcome. While you can't change the opponent, you can study their strategies and weaknesses to gain a competitive edge. For offshore matches, understanding and adapting to the local conditions (like weather, pitch, etc.) can be beneficial.
4. **Season (93.93)**: The time of the year when the match is played (season) also has a significant relationship with the outcome. This could be due to factors like weather conditions and player availability. Planning and preparing for these seasonal factors could improve performance.
5. **Match_light_type (45.67)**, **Match_format (21.23)**, **Bowlers_in_team (6.10)**, **All_rounder_in_team (39.71)**, and **First_selection (2.55)**: These variables have lower Chi-square scores, but they still affect the match outcome. Adjusting the team composition (like the number of bowlers or all-rounders) and the batting order (first selection) based on the match format and light type could be beneficial.

Remember, these insights are based on statistical relationships and do not guarantee a win. Cricket is an unpredictable game, and the outcome can be influenced by many factors. However, understanding these relationships and making data-driven decisions can certainly help improve the team's performance.

Match Format: 1

Model	Train Accuracy	Test Accuracy	Train F1-score (Class 0)	Train F1-score (Class 1)	Test F1-score (Class 0)	Test F1-score (Class 1)
Random Forest	1.00	0.95	1.00	1.00	0.86	0.97
Neural Network	0.79	0.79	0.00	0.88	0.00	0.88

Match Format: 2

Model	Train Accuracy	Test Accuracy	Train F1-score (Class 0)	Train F1-score (Class 1)	Test F1-score (Class 0)	Test F1-score (Class 1)
KNN	0.83	0.73	0.00	0.91	0.00	0.84
Decision Tree	1.00	0.97	1.00	1.00	0.95	0.98

Fig - 28 Performance of each model with match format as a Target variable

Match Format: 1

- **Random Forest**:
    - Training: Perfect performance with 100% accuracy, precision, recall, and F1-score for both classes.
    - Testing: Excellent performance with 95% accuracy, high precision and recall for both classes, and F1-scores above 0.9.
- **Neural Network**:
    - Training: Inferior performance with 79% accuracy, low precision and recall for class 0, and moderate performance for class 1.
    - Testing: Similar inferior performance with 79% accuracy, low precision and recall for class 0, and moderate performance for class 1.

Match Format: 2

- **KNN**:
    - Training: Moderate performance with 83% accuracy, low precision and recall for class 0, and superior performance for class 1.
    - Testing: Similar moderate performance with 73% accuracy, low precision and recall for class 0, and good performance for class 1.
- **Decision Tree**:
    - Training: Perfect performance with 100% accuracy, precision, recall, and F1-score for both classes.
    - Testing: Excellent performance with 97% accuracy, high precision and recall for both classes, and F1-scores above 0.95.

Insights and Recommendations

Based on the performance metrics, here are some insights and recommendations:

- **Random Forest** is the best model for Match Format: 1, with excellent performance on both training and testing datasets.
- **Decision Tree** is the best model for Match Format: 2, with perfect performance on the training dataset and excellent performance on the testing dataset.
- **Neural Network** performs poorly on both match formats, indicating that it may not be the best choice for this classification task.
- **KNN** performs moderately on Match Format: 2, but its performance is not as good as the Decision Tree model.

The reasons for the inferior performance of the Neural Network model are not immediately clear from the classification reports. However, some reasons could be:

- Overfitting: The model may be overfitting to the training data, resulting in mediocre performance on the testing dataset.
- Insufficient training data: The model may require more training data to learn the patterns and relationships in the data.

Strategic Recommendations:

1. **Balanced Team Composition**:
    - Ensure a mix of experienced players and young talents.
    - Include multiple all-rounders for flexibility.
2. **Special Focus on Key Players**:
    - Ensure top batsmen and bowlers are in form and fit.
    - Focus on key players' mental and physical preparation.
    - Give preference to all-rounders & fast bowlers.
3. **Match-specific Preparation**:
    - Tailor strategies and player selection based on the opponent and match conditions.
    - Simulate match conditions during practice sessions.
    - Take each match as an opportunity to change team dynamics.
4. **Reduce Extras**:
    - Implement strict bowling discipline to minimize extra bowls.
    - Capitalize on opponent’s mistakes by taking advantage of extra runs.
    - Start regular intensive bowling camps with training from experts.
5. **Audience Engagement**:
    - Use high audience matches to boost team morale and energy.
    - Engage with the crowd to create a supportive atmosphere.
    - This does not impact win or loss, but home ground advantage cannot be neglected, and we need to host more matches for that to improve.

Match 1: 1 Test match with England in England

Initial Prediction: Loss

Reason: England's home advantage, rainy conditions, and India's relatively weaker performance in Test matches in England.

Suggested Changes:

1. **Team Composition:** Include an extra spinner in the team, considering the rainy conditions and England's weakness against spin.
2. **Batting Strategy:** Focus on playing defensively, with a higher emphasis on survival rather than scoring quickly. This will help India build partnerships and wear down the English bowlers.
3. **Bowling Strategy:** Use the conditions to India's advantage by employing a more aggressive bowling approach, with a focus on swing and seam bowling.

Match 2: 2 T20 matches with Australia in India

Initial Prediction: Loss

Reason: Australia's strong T20 team, India's relatively weaker performance in T20s against Australia, and the home advantage not being significant enough.

Suggested Changes:

1. **Team Composition:** Include an extra all-rounder in the team, considering the importance of power-hitting and bowling variations in T20s.
2. **Batting Strategy:** Focus on playing aggressively, with a higher emphasis on scoring quickly and taking advantage of the powerplay overs.
3. **Bowling Strategy:** Employ a more varied bowling attack, with a mix of pace, spin, and slower balls to keep the Australian batsmen guessing.

Match 3: 2 ODI matches with Sri Lanka in India

Initial Prediction: Loss

Reason: Sri Lanka's recent improvements in ODI cricket, India's weaker performance in ODIs against Sri Lanka, and the home advantage not being significant enough.

Suggested Changes:

1. **Team Composition:** Include an extra batsman in the team, considering the importance of building partnerships and scoring big in ODI’s